SATURATION OPERATIONS ON MULTIPLICATIVE SYSTEMS AND SUBCATEGORIES ARE MUTUALLY INVERSE

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Editorial note. Counted span is the author's own proof environment, source lines 2413-2445, following the statement at 2394-2411.

Editorial note. No omissions: both identifications are displayed as chained equalities with named lemmas.

Allowed standard prerequisites (editorial).

- Kernel of the quotient. lemma-kernel-quotient (same chapter, cited at ~2420). Identifies B(S(B)) with the kernel of the quotient, giving (2).
- Arrows inverted by the localization. lemma-third-object-zero (same chapter, used in the localized category) and Categories, Lemma categories-lemma-what-gets-inverted (cited at ~2440). Identifies S(B(S)) with the set of arrows inverted by Q, and the final statement of that lemma closes (1).

Source reference. Original reference 05Q9: focused-seeds/geometry/S9-der-operations/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05RK: sources/stacks-derived/pinned-original-source.tex, source lines 2338–2338. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QR: sources/stacks-derived/pinned-original-source.tex, source lines 833–833. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. The operations described above have the following properties*

(1) *$S(\mathcal{B}(S))$ is the "saturation" of S, i.e., it is the smallest saturated multiplicative system in $\mathcal{D}$ containing S, and*
(2) *$\mathcal{B}(S(\mathcal{B}))$ is the "saturation" of $\mathcal{B}$, i.e., it is the smallest strictly full saturated triangulated subcategory of $\mathcal{D}$ containing $\mathcal{B}$.*

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

In particular, the constructions define mutually inverse maps between the (partially ordered) set of saturated multiplicative systems in $\mathcal{D}$ compatible with the triangulated structure on $\mathcal{D}$ and the (partially ordered) set of strictly full saturated triangulated subcategories of $\mathcal{D}$.

Proof. First, let's start with a full triangulated subcategory $\mathcal{B}$. Then $\mathcal{B}(S(\mathcal{B})) = \mathrm{Ker}(Q : \mathcal{D} \to \mathcal{D}/\mathcal{B})$ and hence (2) is the content of Lemma 05RK.

Next, suppose that S is multiplicative system in $\mathcal{D}$ compatible with the triangulation on $\mathcal{D}$. Then $\mathcal{B}(S) = \mathrm{Ker}(Q : \mathcal{D} \to S^{-1}\mathcal{D})$. Hence (using Lemma 05QR in the localized category)

$$\begin{aligned} S(\mathcal{B}(S)) &= \left\{ \begin{array}{c} f \in \mathrm{Arrows}(\mathcal{D}) \text{ such that there exists a distinguished} \\ \text{triangle } (X, Y, Z, f, g, h) \text{ of } \mathcal{D} \text{ with } Q(Z) = 0 \end{array} \right\}. \\ &= \{f \in \mathrm{Arrows}(\mathcal{D}) \mid Q(f) \text{ is an isomorphism}\} \\ &= \hat{S} = S' \end{aligned}$$

in the notation of Categories, Lemma 05Q9. The final statement of that lemma finishes the proof. □